

```
In [9]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [13]: df = pd.read_csv("titanic.csv")
```

```
In [14]: df.head()
```

```
Out[14]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



```
In [15]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [16]: df.describe()
```

Out[16]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [17]: df.isnull().sum()
```

```
Out[17]: PassengerId    0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                 177
SibSp                0
Parch                0
Ticket              0
Fare                 0
Cabin               687
Embarked             2
dtype: int64
```

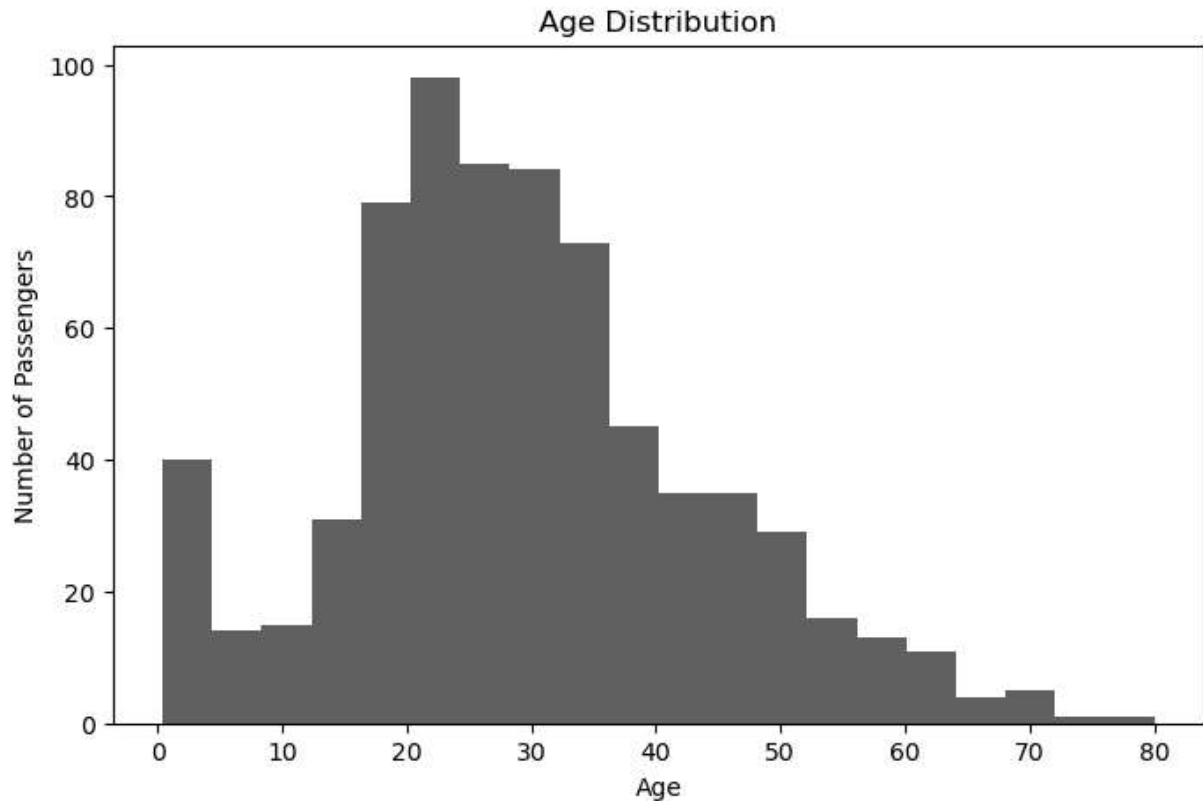
```
In [18]: df['Sex'].value_counts()
```

```
Out[18]: Sex
male      577
female    314
Name: count, dtype: int64
```

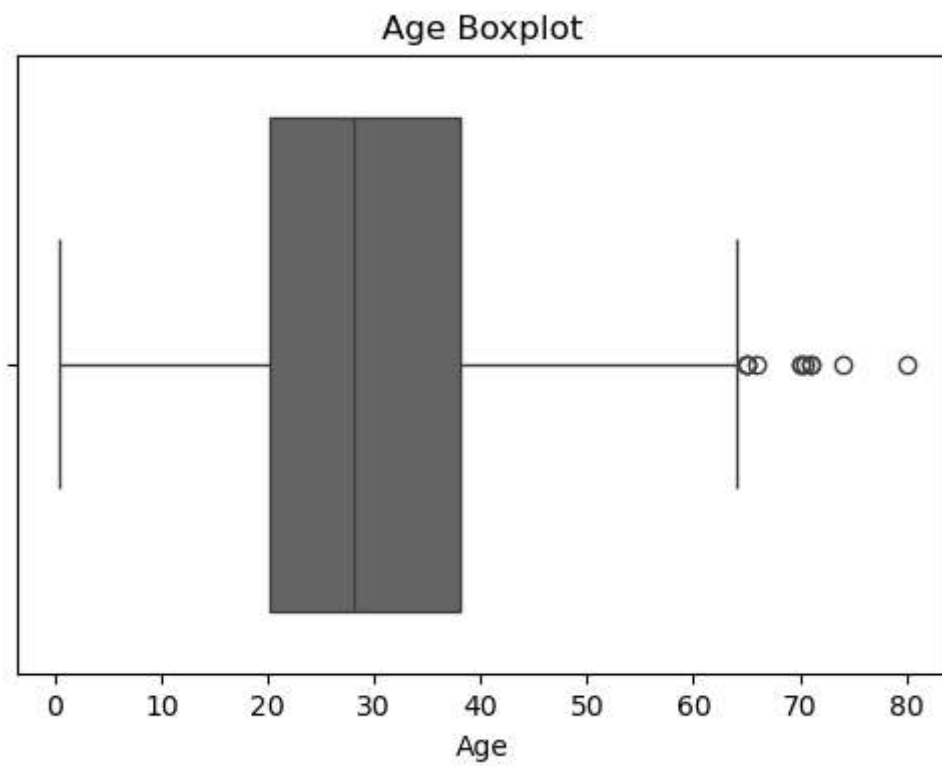
```
In [19]: df['Survived'].value_counts()
```

```
Out[19]: Survived
0      549
1      342
Name: count, dtype: int64
```

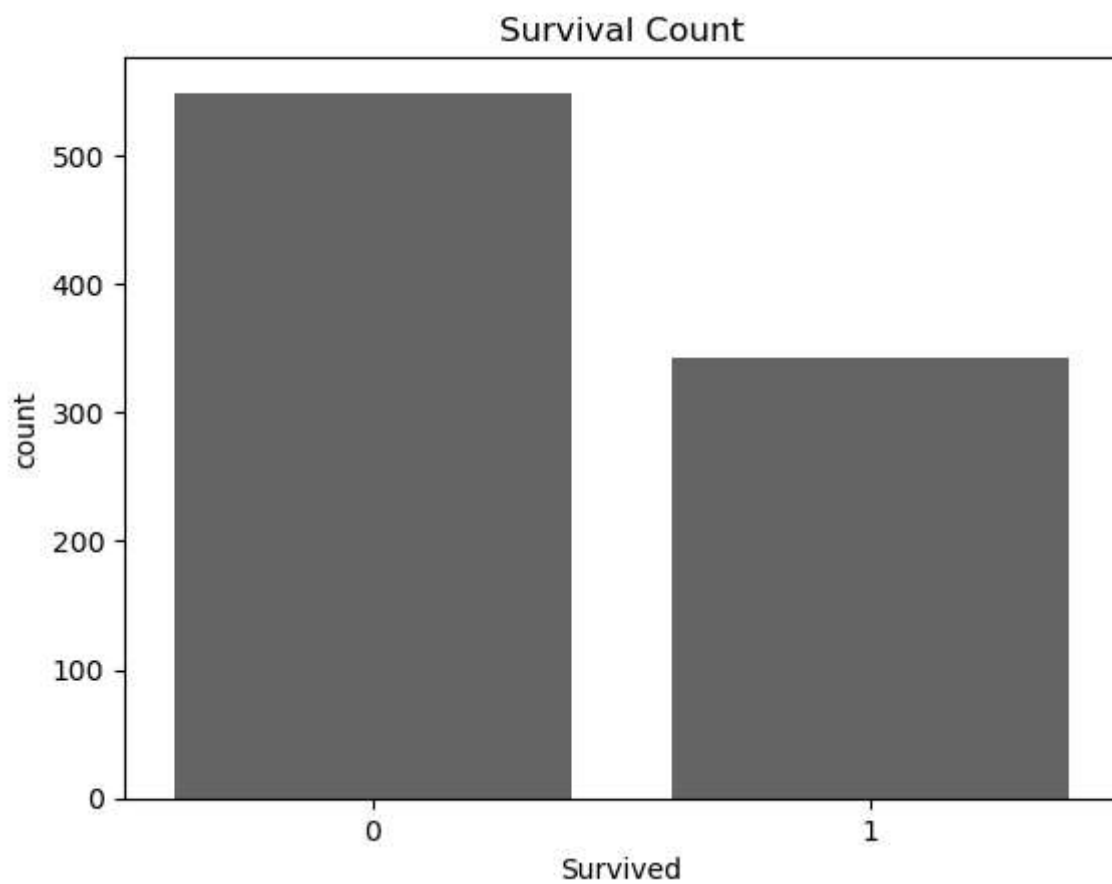
```
In [20]: plt.figure(figsize=(8,5))
plt.hist(df['Age'].dropna(), bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



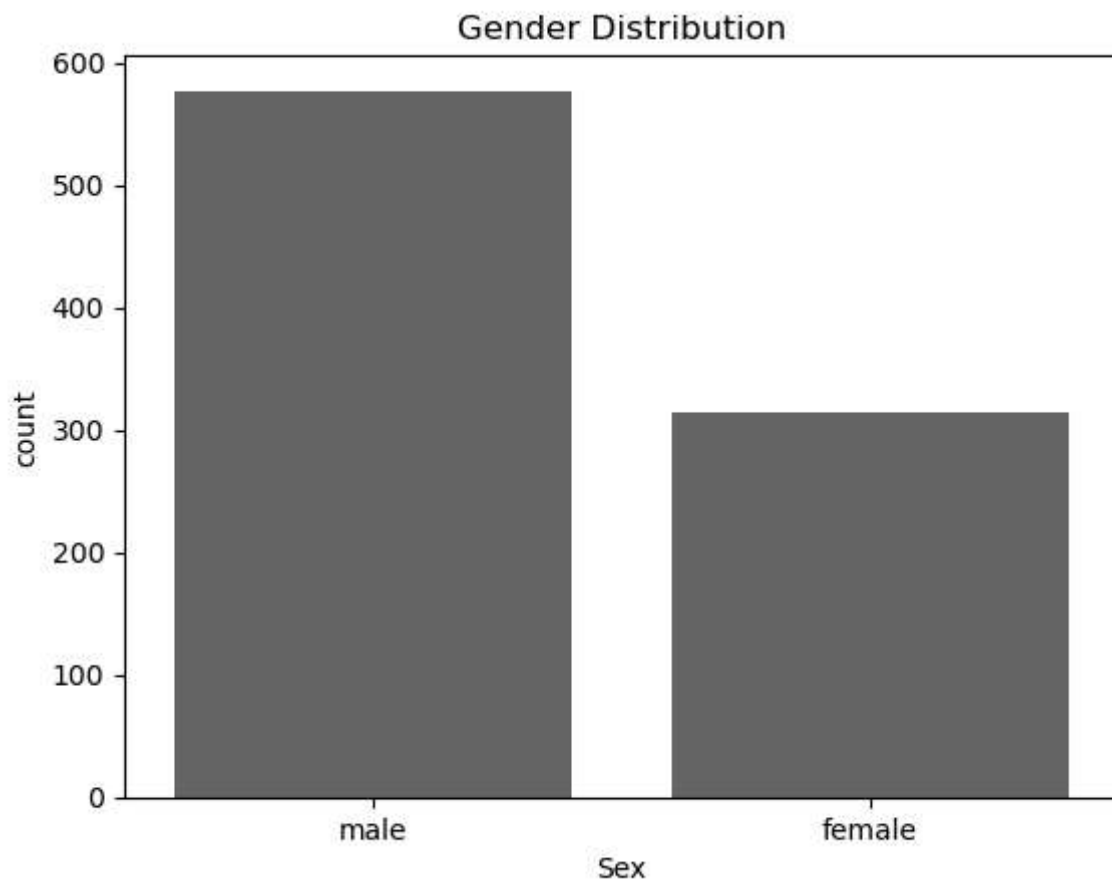
```
In [21]: plt.figure(figsize=(6,4))
sns.boxplot(x=df['Age'])
plt.title("Age Boxplot")
plt.show()
```



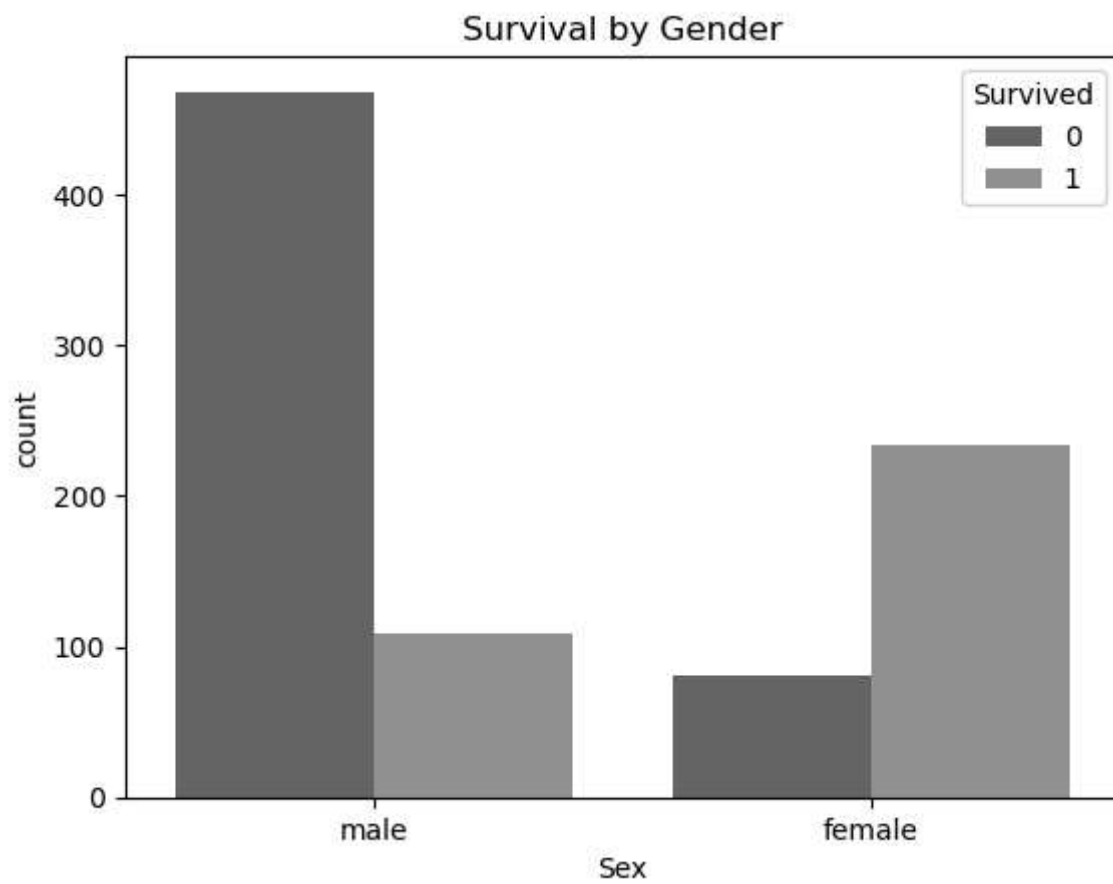
```
In [22]: sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



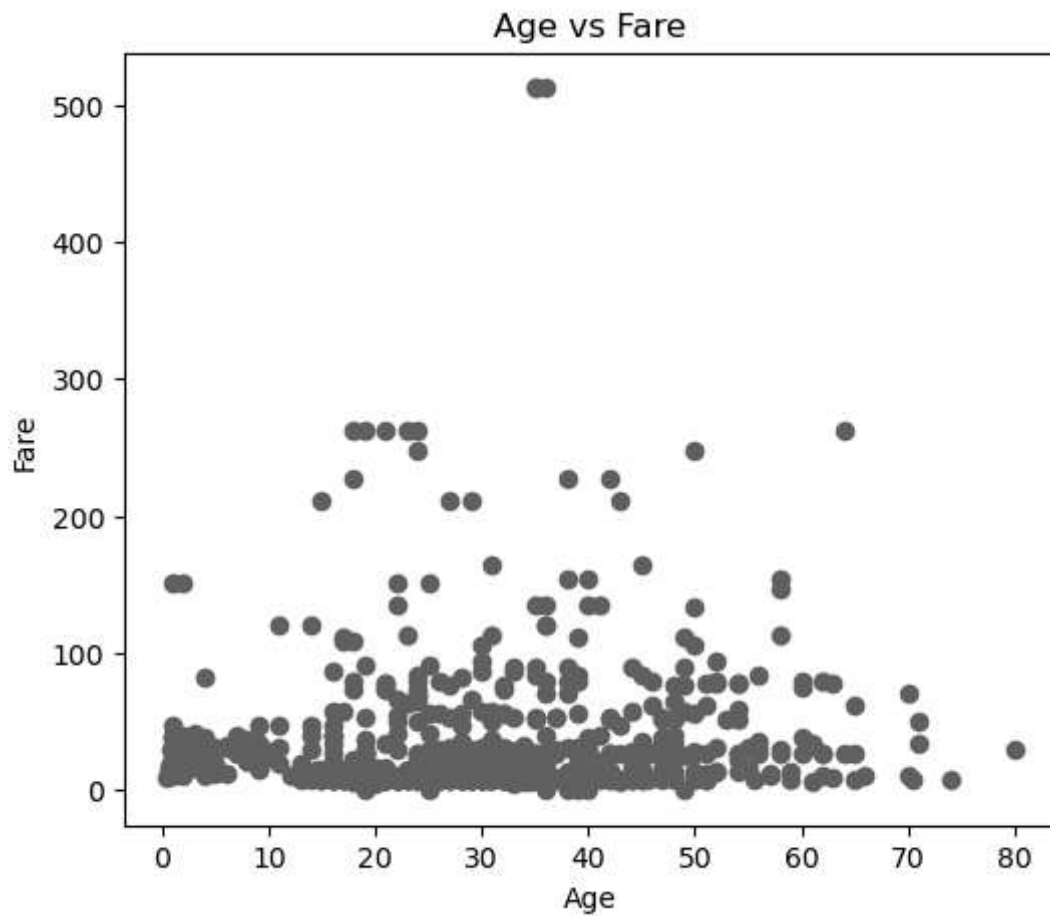
```
In [23]: sns.countplot(x='Sex', data=df)
plt.title("Gender Distribution")
plt.show()
```



```
In [24]: sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()
```

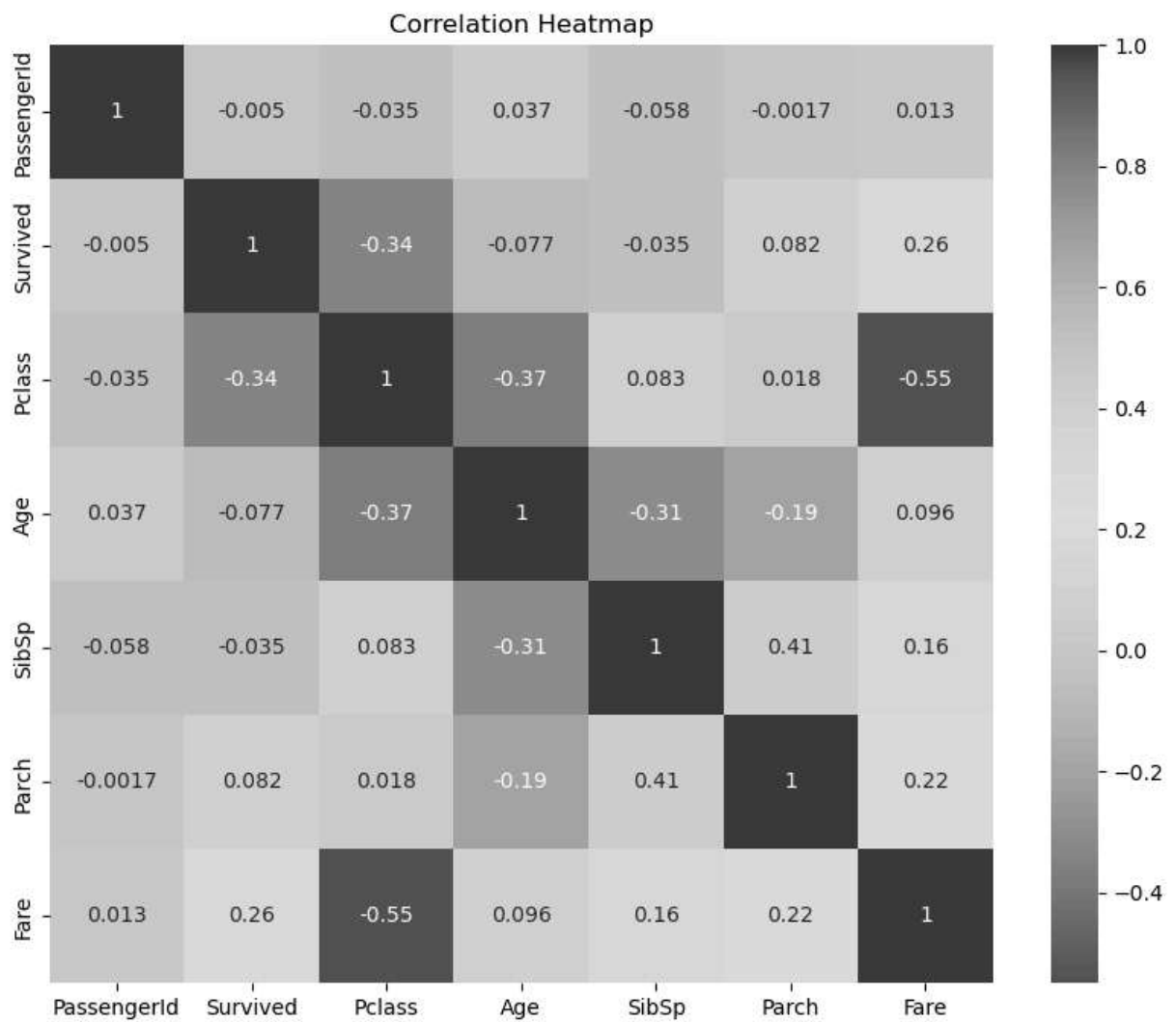


```
In [25]: plt.figure(figsize=(6,5))  
plt.scatter(df['Age'], df['Fare'])  
plt.xlabel("Age")  
plt.ylabel("Fare")  
plt.title("Age vs Fare")  
plt.show()
```

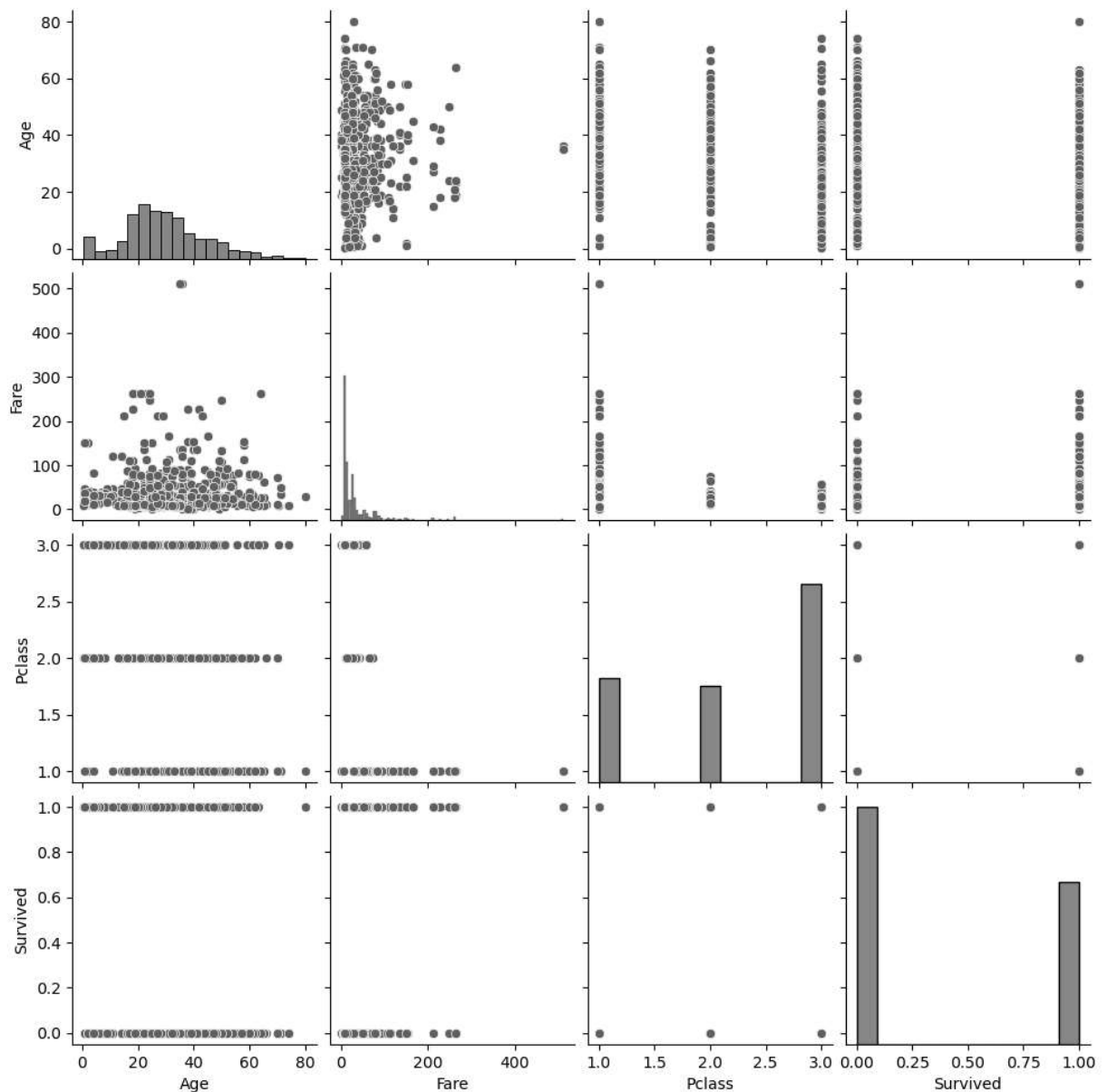


```
In [26]: numeric_df = df.select_dtypes(include=['number'])
```

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In [27]: plt.figure(figsize=(10,8))  
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')  
plt.title("Correlation Heatmap")  
plt.show()
```



```
In [28]: sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']].dropna())  
plt.show()
```



```
In [ ]: The dataset contains 891 passengers.
Some columns have missing values.
Most passengers were between 20-40 years old.
Male passengers were more than female passengers.
More passengers died than survived.
Female passengers had a better survival rate.
Age contains some outliers.
Fare varies significantly among passengers.
Heatmap shows weak correlations between most variables.
Pairplot helps visualize relationships among numeric features.
```